

**Dare to build the next generation of
realtime translation devices powered
by Edge AI**

Presented by Saigon AI Hub and Qualcomm

TECHNICAL PROPOSAL

Phase 2 — Technical Submission

Team / Project Name	Say It For Me
Submission Date	21 / 08 / 2026
Version	v1.0
Confidentiality	Restricted — Challenge Review Only

Evaluation Criteria & Scoring Rubric

Criterion	Weight	What Judges Look For
1. AI Approach & Technical Design	35%	Pipeline completeness, model choices, on-device optimisation strategy, accuracy targets and evaluation plan
2. Hardware & Device Concept	25%	Platform justification, BOM feasibility, power budget, form-factor suitability for the target environment
3. Business Solution	15%	How the solution addresses the industry gap, differentiation from existing tools, innovation strength, real-world applicability
4. Problem Definition & Impact	15%	Clarity of problem, real-world relevance, target user insight, design constraints identified
5. Team and Execution Plan	10%	Complementary skills, realistic timeline, milestone feasibility Early demo is a plus
TOTAL	100%	

Evidence labels used throughout: IMPLEMENTED = code exists; MEASURED = reproducible observed result; TARGET/PROPOSED = future design; LIMITATION = not yet verified.

1. Executive Summary

Evidence labels: IMPLEMENTED = code/test contract; TARGET/PROPOSED = future work; LIMITATION = unverified.

1.1 Problem Overview

Vietnamese-speaking production operators and Korean-speaking supervisors, engineers and trainers need short, clear exchanges during operations, maintenance, QA and safety work. Noise, gloves and time pressure make typing inconvenient; intermittent or restricted connectivity and operational terminology further limit a generic cloud-first workflow.

1.2 Proposed Solution

Say It For Me is an offline-first Vietnamese↔Korean voice-translation architecture designed for smartphone deployment. Its Phase 2 reference path makes language direction explicit, validates audio input, converts Vietnamese speech to text with a real Whisper adapter, and translates VI→KO with real NMT adapters; Korean TTS is proposed for Phase 3.

1.3 Key Value Proposition

The differentiator is the proposed FactorySafe Translation Guard: explicit direction, protected technical tokens, visible transcript/translation before audio, and a block/repeat path when validation fails. The repository implements typed direction contracts, a local model lifecycle foundation, Whisper/NMT adapters, a reproducible CLI and an evaluator. M2M100-418M is the provisional MIT-license baseline. The CLI has no mock fallback; no successful real-audio ASR→NMT run is claimed. Phase 3 will test device execution, privacy behavior and reviewer-approved Korean terminology against held-out factory scenarios before any production use. Until then, outputs remain clearly decision support for low-risk exchanges, with visible evidence labels separating implemented behavior from targets and proposals.

2. Problem Definition & Target Users

↗ Scoring weight: 15% — Problem Definition & Impact

2.1 Problem Statement

Routine Vietnamese↔Korean factory communication can involve machine status, warnings, measurements and maintenance instructions. Misheard or delayed clarification can increase operational risk. Say It For Me is intended for short, lower-risk interactions alongside established safety procedures and human escalation; it does not replace certified interpretation or safety controls.

2.2 Impact Analysis

Impact Area	Current Pain Point	Consequence
Safety / Operations	<i>Misunderstood machine state, warning or operating instruction</i>	<i>Increased operational/safety risk; clarification is required</i>
Productivity	<i>Repeated clarification and manual relay</i>	<i>Slower coordination and possible rework</i>
Connectivity	<i>Restricted or weak-network zones</i>	<i>Cloud-dependent translation may be unavailable or unsuitable</i>

2.3 Target Users & Use Cases

User Segment	Language Need	Primary Context	Priority
<i>Vietnamese production operators</i>	<i>VI ↔ KO</i>	<i>Assembly and machine operations</i>	<i>Critical</i>
<i>Korean supervisors / engineers</i>	<i>KO ↔ VI</i>	<i>Production coordination / maintenance</i>	<i>Critical</i>
<i>Maintenance / QA / technical trainers</i>	<i>VI ↔ KO</i>	<i>Maintenance, QA and training</i>	<i>High</i>

2.4 Key Design Constraints

Constraint	Target / Requirement
Internet dependency	<i>TARGET — no cloud inference at runtime after provisioning; network-disabled verification pending</i>
End-to-end translation latency	<i>TARGET — under 3 seconds after utterance end; not measured end-to-end</i>
Target deployment environment	<i>Industrial smartphone / factory floor</i>
Language pairs required	<i>VI ↔ KO; current real implementation priority: VI → KO</i>
Additional constraints	<i>Hands-busy interaction, noisy environment, protected technical tokens and local-processing/privacy target</i>

3. Business Solution & Innovation

↗ Scoring weight: 15% — Business Solutions

3.1 Industry Problem & Solution Fit

Generic cloud translation apps are useful general-purpose tools but may rely on connectivity/cloud processing and are not designed around factory-safe technical-token validation. Human bilingual intermediaries are high-value for complex communication but may not be instantly available for every routine exchange. Static phrasebooks are

predictable for known phrases but cannot easily cover unexpected machine states or free-form descriptions. Say It For Me targets the gap with a VI↔KO manufacturing focus, an offline-first smartphone architecture, explicit direction, and proposed protected-token validation.

3.2 Innovation & Competitive Strengths

Dimension	Existing Solutions	Your Solution
Connectivity	<i>Connectivity may be required depending on the tool</i>	<i>TARGET: local inference after provisioning; offline verification pending</i>
Latency	<i>Varies by solution and network</i>	<i>TARGET: under 3 seconds; not measured end-to-end</i>
Domain Accuracy	<i>General-purpose vocabulary</i>	<i>Factory/domain evaluation and FactorySafe token design; Korean review pending</i>
Data Privacy	<i>Some cloud services transmit data externally</i>	<i>Architecture targets local processing; network-disabled verification pending</i>

FactorySafe status: proposed validation layer; no production safety claim.

FactorySafe is a proposed safety-oriented layer rather than an accuracy claim. It is designed to preserve technical identifiers, show text before playback and request repetition when a validation rule fails. Current Phase 2 evidence supports the reference implementation and test contracts, not production safety validation.

4. AI Approach & Technical Design

↗ Scoring weight: 35% — Highest weighted criterion

4.1 System Pipeline Overview

IMPLEMENTED reference path: mono 16 kHz PCM16 WAV → validation/segmentation → Whisper Tiny via faster-whisper/CTranslate2 → explicit Vietnamese text → VI→KO NMT adapter → Korean text. TTS and handset delivery are PROPOSED Phase 3 stages. Each transition carries source/target language and diagnostics; the strict CLI rejects unsupported input rather than fabricating output.



4.2 Module-by-Module Design

4.3 On-Device Optimisation & Memory Management

4.4 Robustness & Edge Case Handling

5. Hardware & Device Concept

↗ Scoring weight: 25% — Hardware & Device Concept

5.1 Platform Selection & Justification

PROPOSED platform comparison: Snapdragon 8 Gen 3 Android class is the Phase 3 target; Snapdragon 7+ Gen 3 Android class is a lower-tier candidate. Both require OEM-specific validation.

Criteria	Option A — Snapdragon 8 Gen 3 Android class	Option B — Snapdragon 7+ Gen 3 Android class	Selected?
NPU Performance (TOPS)	<i>Vendor-advertised; verify on handset</i>	<i>Vendor-advertised; verify on handset</i>	Proposed: Option A; no benchmark
Power Consumption (TDP)	<i>Not measured</i>	<i>Not measured</i>	Proposed: Option A; no benchmark
RAM / Storage Support	<i>OEM-specific; not selected</i>	<i>OEM-specific; not selected</i>	Proposed: Option A; no benchmark
AI SDK / Toolchain	<i>QAI RT/QNN candidate</i>	<i>QAI RT/QNN candidate</i>	Proposed: Option A; no benchmark
Form Factor Suitability	<i>Smartphone class</i>	<i>Smartphone class</i>	Proposed: Option A; no benchmark

PROPOSED selection: a Snapdragon 8 Gen 3 Android-class phone is the Phase 3 feasibility target because it offers a current mobile AI platform and Qualcomm toolchain route. No handset has been selected or benchmarked; dimensions, weight, environmental rating, RAM, runtime and thermal behavior remain device-dependent verification work.

5.2 Key Hardware Components & Power Budget

Component	Specification	Peak Power	Notes
Mobile SoC	<i>Proposed Snapdragon 8 Gen 3 Android class</i>	<i>Not measured</i>	<i>PROPOSED; no team handset</i>
Phone microphone	<i>Built-in mic; 16 kHz ASR input after preprocessing</i>	<i>Not measured</i>	<i>OEM audio path; verify on handset</i>
Phone speaker / audio out	<i>Built-in speaker; wired/BT if needed</i>	<i>Not measured</i>	Playback path to validate
Battery	<i>Integrated smartphone battery</i>	—	<i>Runtime not measured</i>
Storage	<i>Local model storage; OEM capacity</i>	<i>Not measured</i>	<i>App and local model assets</i>
Display (optional)	<i>Phone touch display</i>	<i>Not measured</i>	UI and transcript display
Total System Budget	Not estimated	<i>Not measured</i>	<i>Measure active, idle and thermal behavior on selected handset</i>

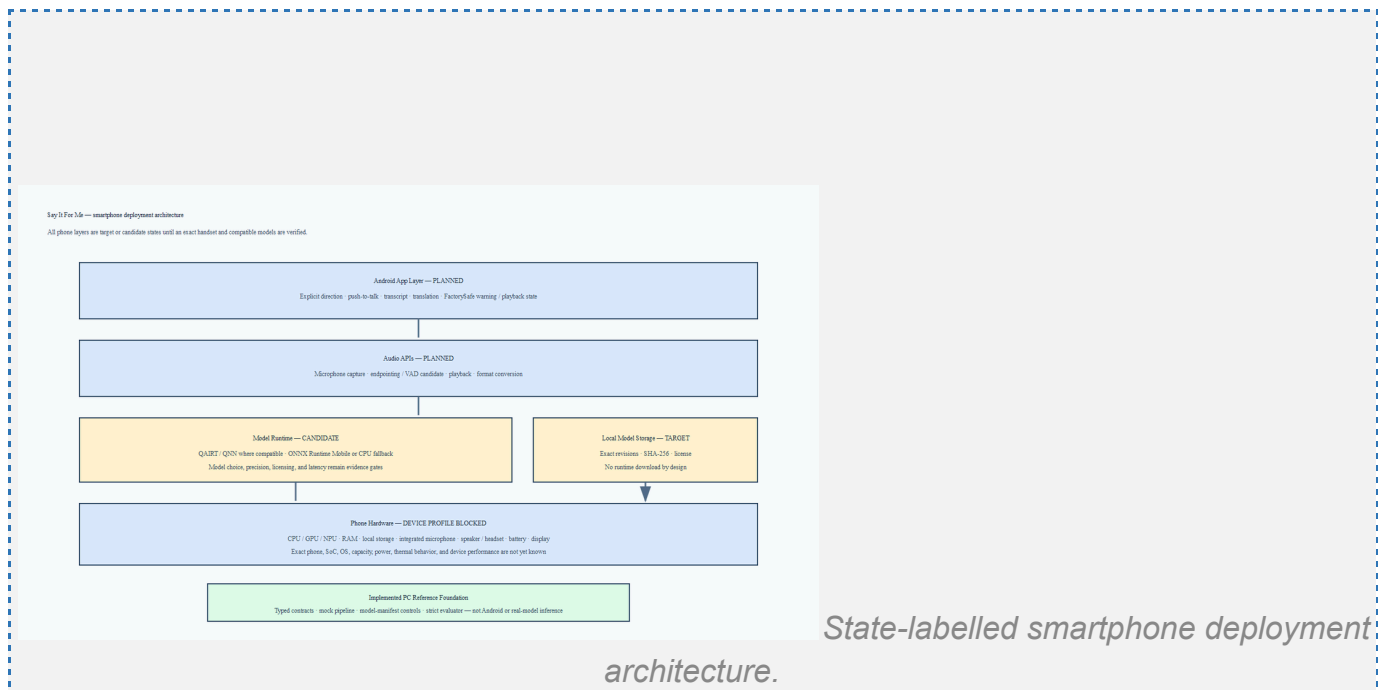
6. System Architecture & Integration

Integration boundary: current Python reference components define the contracts; Android, QAIRT/QNN and handset audio I/O are proposed Phase 3 integration targets.

6.1 Software Stack

Layer	Component / Framework	Role
OS	<i>CURRENT: desktop Python;</i> <i>PROPOSED: Android</i>	Base runtime & audio I/O
AI Runtime	CTranslate2 ASR; Transformers/PyTorch NMT; QAIQT/QNN proposed	NPU inference execution
Audio Processing	Typed audio contracts + bounded segmentation; production VAD proposed	Noise reduction & VAD
Model Serving	Local lifecycle/lazy loading; mobile runtime proposed	ASR, NMT, TTS inference
App Layer	Python CLI/pipeline; Android client proposed	UI, pipeline orchestration

6.2 Smartphone Deployment Architecture



6.3 Offline-First Design Principles (if any)

1. **TARGET:** provision ASR/NMT assets locally; network-disabled operation has not been verified.
2. **IMPLEMENTED:** model manifest/checksum and lazy loading provide a foundation for local assets; VI→KO is the reference direction.

3. *PROPOSED*: show transcript and translation before optional Korean playback; do not treat cached text as a safety decision.
4. *LIMITATION*: Korean TTS, a selected phone, and an offline end-to-end run are not implemented or measured in Phase 2.

7. Team Profile & Project Timeline

↗ Scoring weight: 10% — Team and Execution Plan

7.1 Team Members

Name	Role	Expertise	Contribution Area
Nguyễn Đồng Thiên Kỳ	Technical Lead (TL)	Architecture, integration, scope control	Technical decisions, integration, final approval
Nguyễn Tiến Đạt	AI/Translation Engineer (ATE)	ASR, VI→KO NMT, model evaluation	ASR/NMT selection and evaluation
Nguyễn Đăng Gia Đạo	Audio & Device Engineer (AUD)	Audio/WAV, TTS, device/hardware	Audio contract, TTS/device feasibility
Hà Duy Lộc	Application & Data Engineer (APP)	Dataset, evaluation, application/demo	Evaluator, application, proposal/submission

7.2 Project Timeline

Phase	Milestone	Key Activities	Target Date
1	Requirements & architecture foundation	Direction contracts, pipeline/evaluator and proposal foundation	Before 21/08/2026
2	Reference pipeline & proposal	Model lifecycle, diagnostics and submission evidence	Before 21/08/2026
3	ASR/NMT integration	Whisper/NMT adapters, strict CLI and tests	21/08/2026
4	Phase 2 submission	Final proposal, PDF export and upload package	21/08/2026

8. Submission Checklist

All items must be completed before final submission. Missing sections score zero on the relevant rubric criterion.

#	Checklist Item	Status
1	Executive Summary written (200–300 words)	[x] Done

2	Problem Statement, Target Users, and Design Constraints completed	[x] Done
3	Business Solution completed — industry gap, solution fit, and competitive differentiation described	[x] Done
4	AI pipeline documented with model specs, latency targets, and optimisation strategy	[x] Done
5	Hardware platform justified; form factor, BOM and power budget filled in	[x] Done
6	Team profiles and project timeline filled in	[x] Done
7	All content completed; pipeline and architecture diagrams inserted	[x] Done
8	Demo video or prototype link attached (if available); document exported as .PDF	[x] Done